

code that's better organized, more compact, and more powerful. Even skilled programmers can benefit from going back to the basics and mastering fundamentals until they're second nature. Many talented programmers write disorganized, badly documented, or inefficient code and think they can get away with it because it “works.” But remember that code doesn't usually succeed on its own—it is almost always part of a broader program, some team effort or grand business project that relies on cooperation between people and over time. Because of this, even soft skills like planning, oral and written communication, negotiation, and team management can improve your chances of success in the world of algorithms.

If you enjoy creating perfectly optimal algorithms and pushing them to their highest efficiency, you're in luck. For a huge number of computer science problems, there is no known efficient algorithm that runs much faster than brute force. In the next section, we sketch a few of these problems and discuss what's so hard about them. If you, dear adventurer, create an algorithm that solves any of these problems quickly, you could have fame, fortune, and worldwide gratitude for the rest of your life. What are we waiting for? Let's look at some of these challenges for the most courageous among us.

Algorithms for the Ambitious

Let's consider a relatively simple problem related to chess. Chess is played on an 8×8 board, and two opponents take turns moving differently styled pieces. One piece, the queen, can move any number of squares along the row, column, or diagonal where it is placed. Usually, a player possesses only one queen, but it's possible for a player to have up to nine queens in a standard chess game. If a player has more than one queen, it may be that two or more queens “attack” each other—in other words, they are placed on the same row, column, or diagonal. The *eight queens puzzle* challenges us to place eight queens on a standard chessboard such that no pair of queens is on the same row, column, or diagonal. [Figure 11-1](#) shows one solution to the eight queens puzzle.